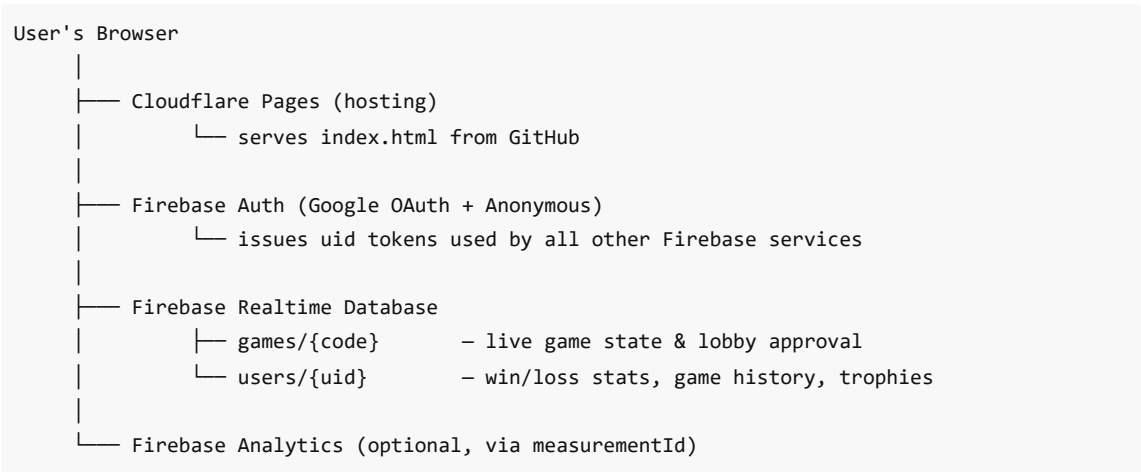


Tactical Chess

A real-time tactical chess app with a built-in coach, online multiplayer, computer AI, and user progress tracking.

Live site: <https://tactical-chess.pages.dev>

Services & How They Connect



1. Cloudflare Pages — Hosting

What it does	Serves the static index.html to users
Connected to	GitHub repo <code>omridunour/tactical-chess</code> (auto-deploys on push to master)
Dashboard	https://dash.cloudflare.com → Pages → tactical-chess
Live URL	https://tactical-chess.pages.dev

Every `git push` to `master` triggers a new Cloudflare Pages build automatically — no manual deploy step needed.

2. Firebase Authentication

What it does	Handles user identity: Google sign-in and anonymous (guest) sessions
Providers enabled	Google OAuth, Anonymous
Connected to	Firebase Realtime Database (rules check <code>auth.uid</code>)
Dashboard	Firebase Console → tactical-chess-c8b9c → Authentication

Authorized domains that must be listed in Firebase Console → Authentication → Settings → Authorized domains:

- `tactical-chess.pages.dev`
- `localhost` (for local development)

Guest mode: Users who click "שחק כאורח" get an anonymous Firebase session. They can play local and online games. After winning, a "שמור חשבון" button lets them link their anonymous session to a real Google account —

preserving their game history.

3. Firebase Realtime Database

What it does	Stores live game state for online multiplayer and user stats
Region	europa-west1
Dashboard	Firebase Console → tactical-chess-c8b9c → Realtime Database

Database structure

```
/games
  /{gameCode}
    board[][]          - current board state
    currentPlayer      - "white" | "black"
    castlingRights     - object
    enPassantTarget    - object | null
    moveHistory[]      - array of move strings
    status             - "waiting" | "active"
    players/
      white/           - { uid, name, photo }
      black/           - { uid, name, photo } (null until approved)
    requests/
      /{uid}           - { name, photo, status: "pending"|"approved"|"rejected" }
    createdAt          - timestamp

/users
  /{uid}
    name              - display name
    email
    photo             - avatar URL
    lastSeen          - timestamp
    stats/
      wins
      losses
      draws
    gameHistory/
      /{pushId}        - { date, result, opponent, moves, reason, mode, hasTrophy }
    trophies/
      /{pushId}        - { id, date, reason, moves }
```

Recommended database rules

```
{
  "rules": {
    "users": {
      "$uid": {
        ".read": "auth != null && auth.uid == $uid",
        ".write": "auth != null && auth.uid == $uid"
      }
    }
  }
}
```

```
    }
  },
  "games": {
    "$gameId": {
      ".read": "auth != null",
      ".write": "auth != null"
    }
  }
}
```

Set these in Firebase Console → Realtime Database → Rules.

Game Modes

Mode	Description
 מקומי	Two players on the same device
 נגד מחשב	Play as white against a minimax AI (depth 1 = easy, 2 = medium, 3 = hard)
 מקוון	Host creates a lobby, shares a 6-character code, approves join requests before the game starts

Local Development

```
# Install wrangler (Cloudflare CLI – optional, for deploying manually)
npm install

# Open in browser
open index.html
```

The app is a single `index.html` file with no build step. Firebase config is hardcoded in the file (it's safe to expose — Firebase security is enforced by database rules and authorized domains, not by keeping the config secret).

Environment Variables

A `.env` file (git-ignored) holds the Firebase config for reference. The values are also hardcoded in `index.html` because the app has no server-side build step.

```
FIREBASE_API_KEY
FIREBASE_AUTH_DOMAIN
FIREBASE_DATABASE_URL
FIREBASE_PROJECT_ID
FIREBASE_STORAGE_BUCKET
FIREBASE_MESSAGING_SENDER_ID
FIREBASE_APP_ID
FIREBASE_MEASUREMENT_ID
```

Note: *Firebase web API keys are not secret — they identify the project, not grant admin access. Security comes from Firebase Auth rules and the authorized domains list.*

Deployment

Push to `master` → Cloudflare Pages auto-deploys within ~60 seconds.

```
git add .  
git commit -m "your message"  
git push
```